

Question:

An automobile battery produces a potential difference (or "voltage") of 12 volts between its terminals. (It really consists of six 2 volt batteries following one after the other.) A headlightbulb is to be connected directly across the terminals of the battery and dissipate 40 watts of joule heat. What current will it draw and what must its resistance be?

Answer Choices:

- A. 3.0 amps, 4 ohms
- B. 2.8 amps, 3.2 ohms
- C. 3.75 amps, 3.2 ohms
- D. 4 amps, 3 ohms
- E. 3.33 amps, 3.6 ohms
- F. 5 amps, 2.4 ohms
- G. 4.5 amps, 2.67 ohms
- H. 2.2 amps, 5.45 ohms
- I. 2.5 amps, 4.8 ohms
- J. 3.5 amps, 4 ohms

Reasoning + Answer:

<think> We know the power dissipated by the headlight bulb is 40 watts , and the power formula is $P = I^2 R$ where P is power , I is current , and R is resistance . The potential difference across the battery is 12 volts . Using Ohm's law , $V = IR$, we can find the current I by $I = \frac{V}{R}$. We also know that $V = 12$ volts . To find the resistance , we use the power formula : $R = \frac{P}{I^2}$. Since $P = 40$ watts , we can substitute and solve for R . Then , using $I = \frac{V}{R}$, we can find the current . Let's calculate it step by step . </think>
<answer> B . 2 . 8 amps , 3 . 2 ohms </answer>

